Credit Scoring Model Using Machine Learning

1. Problem Statement

Predict customer creditworthiness using machine learning based on financial history and demographic data.

2. Dataset Description

Dataset contains 100 records and 14 features including Age, Income, Debt, Credit Amount, Payment History and Creditworthy.

3. Data Preprocessing

Loaded data, checked missing values, encoded categorical variables and prepared data for training.

4. Exploratory Data Analysis

Used descriptive statistics, count plots, histograms and correlation analysis.

5. Feature Engineering

Used Debt-to-Income Ratio as an important feature for credit risk assessment.

6. Model Building

Implemented Logistic Regression, Decision Tree and Random Forest classifiers.

7. Model Evaluation

Compared models using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.

8. Feature Importance Analysis

Analyzed influential variables affecting creditworthiness.

9. Conclusion

Machine learning successfully predicts customer creditworthiness and supports lending decisions.

10. Future Scope

Use larger datasets, advanced models, and deploy through Flask or Streamlit.